

Encryption key: 12345
Link: <https://github.com/B116564-2022/ICA2resit>

OQSIL

Introduction.

User manual for implementing OQSIL - Online protein Quality aSsesIng tooL into the biological research.

OQSIL - is a generic python based tool, which allows the biologists and other protein focused specialists, the opportunity to find and download the protein fastas from NCBI for specific taxon or organism, conduct quantitative and qualitative analysis: starting from aminoacid count and sequence alignment and ending with identifying plotting conserved regions, calculating similarity scores, identifying motifs and physicochemical features of individual proteins.

1. USER MANUAL

PREREQUISITES

The program is designed to be utilised on MSC8 server, please make sure, that You are on this server and the program file is in your current folder. If you dont have access this server, to start working with this tool, please make sure, that your system has installed Entrez utilities, Clustalo and EMBOSS softwares. (Small hint: they might be easily found googling) and the current tools is in the same directory as You.

SUBSEQUENT ANALYSIS:

- 1.1. Go to <https://github.com/B116564-2022/ICA2resit>
- 1.2. Download the file B116564-2022.ICA2resit.tar.gz.cpt into Your current working directory
- 1.3 Type
ccdecrypt B116564-2022.ICA2resit.tar.gz.cpt(cc key: 12345)
tar -xvf B116564-2022.ICA2resit.tar.gz.cpt
Now You can start working.
- 1.4. Type ls to see what has been extracted. Make sure oqsil.py file is there
- 1.5. Type p3 resit2.py (if your systems calls python with our name, than use it instead of p3). Greeting from OQSIL should appear here
- 1.6. User enter the name of protein of interest and taxon id

Overall:835 dehydrogenase fastas found.(If the results return zero then user is prompted back to the beginning)
Download all? (y/n):

1.6.1.Type "y" to download all

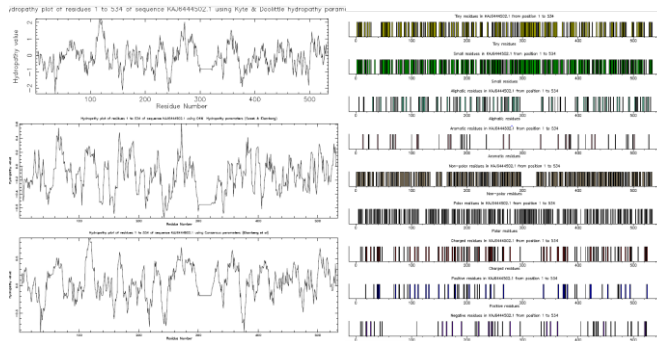
All 83 dehydrogenase fastas for *Aspergillus flavus* have been downloaded to **dehydrogenase.fa**
The min. length of these fastas is 54 bp and
The longest sequence comprises of 1179 bp.
Mean length of dehydrogenase sequences is 371.9892215568862 bp (Quite a wide range of length)
Select what to do next: (a) Look for another protein(You will be prompted to the beginning), (b) Continue for similarity analysis (c) Exit

1.6.1.1. Type "b" for further similarity analysis, which produces alignment profiles for all the submitted files along with distance matrix. (Note:It might take long if the number of samples is high)

```
Clustal Omega alignment is finished. Generating the results. Please wait
Similarity scores between fastas: [] 835
KAJ6444502.1          0.000000 0.885154 0.872659 0.882022 .....
!!AA_MULTIPLE_ALIGNMENT 1.0
MSF: 3125  Type: P  April 06, 2023 01:37  Check: 6160  ..
Name: KAJ6444502_1          Len:   3125  Check:  185  Weight: -
1.00                                1
50
KAJ6444502_1          ~~~~~ ~~~~~ ~~~~~ ~~~~~ ~~~~~
```


Would You like to analyse the chemical properties of your proteins as well? (y/n):
Type "y" to proceed and enter the name of the fasta file you want to analyse.

Pepinfo finished successfully.
The results saved into pepinfo.pdf



Note. This information is valuable for biologists, as the physicochemical properties of individual proteins, their hydrophobicity might help to identify the protein localisation within or outside the cell. In addition, it is helpful in identifying protein ligand binding regions, and thus in biological modelling and for drug design and development purposes. Personally, I am interested in this property of the biological molecules, as maybe we could distinguish proteome profiles in patients and controls just analysing the aa features of specific proteins, and maybe reveal other underlying mechanisms to incurable diseases. From these graphs we can see that the first protein in our subset presented moderate hydrophobic qualities, and the ration between polar and non polar aa was almost equal. This doesnt allow us to create a valid hypothesis about the localisation of this protein, but leads us to another study. However, this might be a foundation for further research in this direction.

1.7. If you want to filter out the data before starting further processing, than type "n" to 7.1. and press enter

1.7.1. Enter organism name: f.e. *Aspergillus flavus*

Number of results found for *Aspergillus flavus*: 14.
Download all? (y/n)

1.7.2. Type "y" to download all

All 14 dehydrogenase fastas for *Aspergillus flavus* have been downloaded to **dehydrogenase.fa**
The min. length of these fastas is 344 bp
The longest sequence comprises of 362 bp.
Mean length of dehydrogenase sequences is 353.5 bp
Select what to do next: (a) Look for another protein(You will be prompted to the beginning), (b) Continue for similarity analysis (c) Exit

1.7.2.1. Type "b" for further similarity analysis. Ref. 1.6.1.1

Clustal Omega alignment is finished.
Generating the results. Please wait
Similarity scores between fastas: 14
KAJ1713714.1 0.000000 0.005525 0.000000 0.000000 0.005525
0.005525 0.889535 0.889535 0.889535 0.893983 0.893983
0.888252 0.888252 0.888252
!!AA_MULTIPLE_ALIGNMENT 1.0
MSF: 481 Type: P April 06, 2023 01:02 Check: 9470 ..
Name: KAJ1713714_1 Len: 481 Check: 7891 Weight: -1.00
Name: RMZ47514_1 Len: 481 Check: 7831 Weight: -1.00
1

```

50
KAJ1713714_1  MGDAALAPKE HLLITFMRDL APDQLEEIKA KFPGLEVSSV
TLKRGEDIPS
RMZ47514_1    MGDAALAPKE HLLITFMRDL APDQLEEIKA KFPGLEVSSV
TLKRGEDIPS
Would you like to save the output? (y/n)

```

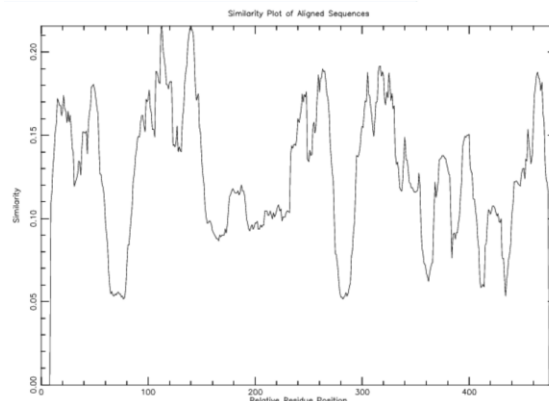
1.7.2.2. Type "y" to save the results

```

Alignment result saved as dehydrogenase.msf.
Similarity scores saved as dehydrogenase.matrix
Do you want to plot the level of conservation between protein
sequences? (y/n): y

```

1.7.2.3. Type "y" Press Enter to plot the level of conservation between protein sequences.



Note: Similarity level for each aminoacid position within aligned protein sequences is visualised here. As the data is sorted for specific organism, in this case we see quite a range of conserved similar regions across the aligned fasta sequences.

1.7.2.4. To exit the plot do not click on x-sign. Press enter and type quit.

```

Conservation plot saved as plotcon.pdf
Do you want to identify protein motifs in your set? (y/n): y

```

1.7.2.5. To identify protein motifs in type "y"

```

# Full: No
# Prune: Yes
# Data_file: /usr/share/EMBOSS/data/PROSITE/prosite.lines
#
=====
Length = 4
Start = position 153 of sequence
End = position 156 of sequence

Motif = AMIDATION
GVHDQVGKRVGILG
  |
 153 156

Length = 28
Start = position 157 of sequence
End = position 184 of sequence

Motif = D_2_HYDROXYACID_DH_1
QVGKRVGILGYGSIQRVSHALGMKVYAYTASPRP
  |                               |
 157                             184

```

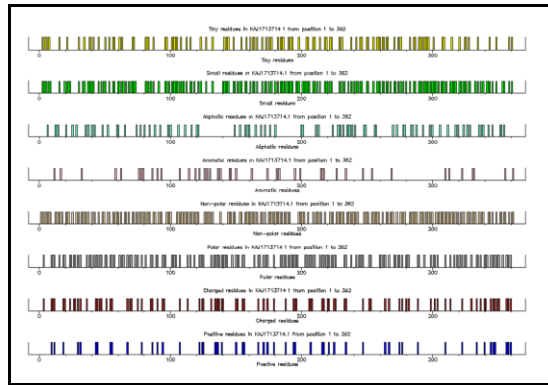
Scan a protein sequence with motifs from the PROSITE database
patmatmotifs finished successfully.
The results saved into **dehydrogenase.patmatmotifs**

1.7.2.6. Would You like to analyse the chemical properties of your proteins as well? (y/n): Type "y" to proceed and enter the name of fasta file which you would like to analyse.

```

Pepinfo finished successfully.
The results saved into pepinfo.pdf

```



2. Developer maintenance manual:

2.1. List of libraries and packages executed while processing this tool

1. Python 3 and above
2. Libraries: os, pandas, subprocess, matplotlib.pyplot, glob
3. Clustal Omega
4. EMBOSS (patmatmotifs vs pepinfo)
5. Ghostscript (for PDF output)

In addition API key was set, at the very beginning to solve the download errors.

2.2. Entrez query.

- a. The code takes a user input protein name(variable:`oq_prot`) and txid(variable:`takson`) number and with this variables conducts the query to Entrez.

`qidiruv_komandasi = f'esearch -db protein -query "{zapros}" | efetch -format fasta'` at the same time calculating the number of found fastas using `os.popen` and `grep` commands

- b. One error trap returns when the number of outcomes is zero, the user is prompted to check the database online and if fastas for this specific query exist, then check the input variables.
- c. When the input is validated and results generated the user might select to process the analysis with the whole set of found data and download it or specify any organism, which they are interested in. downloading files is performed using `os.system` command. And simultaneously mean, max and min length of downloaded fasta sequences is calculated, implementing pandas dataframe.
- d. User is then subsequently prompted for further ClustalO, Prosite Scan and Pepinfo analysis.

2.3. Clustalo analysis and conservation plotting

- a. Filtered or nonfiltered on previous stage fastas are further processed are now assigned to a variable `fasta_fayli` for subsequent alignment and similarity score calculation using ClustalO with `os` and `subprocess` commands.

`clustalo_run = f"clustalo -v -i {fasta_fayli} -o {clust_fayli} --outfmt=msf --threads=20 --force --full --distmat-out={d_file}"`

`try:`

`subprocess.run(clustalo_run, shell=True, check=True, capture_output=True, text=True).`

- b. The results then are visualised on screen with `os`, `gs` and `plt.show()` commands, after which the user is given an option to save the alignment results and distance matrix, which are saved as `msf` and `dist` files.

2.4. Patmatmotifs

- a. To identify matches in the users subset of fastas with Prosite db, the user is then given option to run `patmatmotifs`. Using the following command `patmatmotifs_komandasi = f"patmatmotifs -sequence {yangi_clustalo} -outfile {oq_prot_patmat} -full"`. This command uses as input `msf` output file, generated with `clustalo` assigned to variable `yangi_clustalo`.

- b. Results are saved as the name of the protein `{oq_prot}.patmatmotifs`.

2.5. Pepinfo.

- a. If the user wants to further research psychochemical proteins of individual fastas, one is prompted further, where the initial protein `fa` files is processed with `for` loop to extract individual fastas into files, and the

user is shown the names of files, which one can select and use as an input variable for pepinfo.
`pepinfo_cmd=f"pepinfo -sequence {user_fasta} -outfile {pep_file} -graph pdf"`
Pepinfo result is saved as in pdf and pepinfo formats.

2.6. Limitations and future plans.

Author do understand that the system is limited by a number of factors, solving which might improve the performance and accuracy of this tool. Therefore in further versions, we aim to further

- a. implement error handling traps and the methods of validating user inputs.
- b. Create more opportunities to sort and filter the generated results and customise the study.
- c. Implement more enhanced bioinformatics algorithms to create an opportunity to work with large scale data
- d. Develop a user friendly GUI interface to make life for wet lab and other noncoding biologists easier.

REFERENCE MATERIALS

1. Sievers F., Higgins D. G. Clustal omega //Current protocols in bioinformatics. - 2014. - T. 48. - №. 1. - C. 3.13. 1-3.13. 16.
2. Mullan L., Judge D. P. EMBOSS-A Sequence Analysis Package //NATO SCIENCE SERIES SUB SERIES I LIFE AND BEHAVIOURAL SCIENCES. - 2005. - T. 368. - C. 74.
3. McWilliam H. et al. Analysis tool web services from the EMBL-EBI //Nucleic acids research. - 2013. - T. 41. - №. W1. - C. W597-W600.
4. Baxevanis A. D. Searching the NCBI databases using Entrez //Current protocols in human genetics. - 2006. - T. 51. - №. 1. - C. 6.10. 1-6.10. 24.